Manel Mili

PhD Candidate — Computational Pathology & Explainable AI · Faculty of Sciences of Monastir / LTIM, University of Monastir, Tunisia

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RESEARCH INTERESTS

Deep learning and explainable AI (XAI) for medical imaging and computational pathology, with a focus on weakly-supervised learning, deep neural network architectures, and interpretability methods for clinical prediction tasks. Broader interests include multimodal data fusion, mechanistic interpretability, and AI-driven decision-support systems in healthcare.

EDUCATION

PhD in Computer Science — Computational Pathology

Faculty of Sciences of Monastir (FSM), Tunisia · 2021 – Present

Thesis: “Prediction of MGMT Status in Gliomas Using Artificial Intelligence.” Laboratory: Medical Technologies and Imaging Laboratory (LTIM – LR12ES06), Faculty of Medicine, University of Monastir. Supervisor: Prof. Asma Ben Abdallah.

MSc in Software Engineering

Higher Institute of Informatics and Mathematics of Monastir (ISIMM), Tunisia · 2017 – 2020

Thesis: “Early Mortality Prediction in Intensive Care Units (ICUs) Using Deep Learning Models.” Laboratory: LTIM – LR12ES06.

BSc in Computer Science

ISIMM, Tunisia · 2012 – 2015

Final-year project: “Distributed Optimization by Particle Testing: Case of MaxCSP.”

RESEARCH EXPERIENCE

Research Intern — Mechanistic Interpretability for Multimodal MGMT Prediction

U2IS Laboratory, ENSTA Paris / Institut Polytechnique de Paris, France · Sep – Dec 2025

- Investigated mechanistic interpretability methods for multimodal deep learning models predicting MGMT promoter status, extending the multi-view XAI line of work developed during the PhD.
- Collaborated with the U2IS team on interpretability benchmarking for transformer-based classifiers on biomedical data.

PhD Researcher — MGMT Status Prediction & Explainable AI in Glioma

LTIM, Faculty of Medicine, University of Monastir · 2021 – Present

- Developed an NdLinear-attention–based deep learning model for MGMT promoter methylation prediction in glioblastoma (CoDIT 2026 (Accepted)).
- Designed a quantitatively audited multi-view XAI framework for MGMT methylation classification from histological and imaging data (ICAART 2026 (Published)).
- Built a weakly-supervised, Vision Transformer–conditioned text-explanation model for glioma histology (WCCI-IJCNN 2026 (Accepted)).
- Developed an automated deep learning pipeline for MGMT methylation status prediction from histological images, benchmarked against subjective visual evaluation (IEEE Conference on Cyberworlds, CW 2023 (Published)).
- Built an ensemble learning pipeline for MGMT status prediction in GBM patients, and an earlier gradient-boosting baseline (AMINA 2024, AMINA 2022 (Published)).
- Deployed a Flask-based web interface for MGMT promoter status prediction from histological images, with image upload and inference pipeline for testing.

Applied Machine Learning Projects

- Alzheimer's Disease Prediction — predictive model for Alzheimer's disease in patients. (Python, Keras, TensorFlow)
- Depression Prediction — predictive model for depression risk in individuals. (Python, Deep Learning, TensorFlow)
- ICU Mortality Prediction — deep learning model for early mortality prediction in ICUs using electronic medical record data; published as a book chapter (IDEAL 2022 — see Publications).
- Drug Sensitivity Prediction — model anticipating which patients may develop drug sensitivity. (Python)
- Liver Cancer Risk Prediction — deep learning model estimating the probability of an individual developing liver cancer. (Deep Learning, Statistical Data Analysis)

- Megaloblastic Anemia Prediction — predictive model for the megaloblastic character of macrocytic anemia; published (Leukemia Research Reports — see Publications).
- Adverse Drug Reaction Prediction — model predicting undesirable or harmful reactions to medicines. (Python, Keras, Pandas, TensorFlow)
- COPD Exacerbator Profiling — predicted the profile of frequent exacerbators from the first consultation using clinical and para-clinical data.
- MGMT Methylation Status Prediction (GBM) multimodal deep learning pipeline combining medical imaging and data fusion (see Research Experience and Publications above).
- MGMT Prediction Web Application — Flask-based interface with Ngrok hosting for testing (see above).
- Nosocomial Infection Risk Prediction — model predicting hospital-acquired infection risk for surgical patients. (AI, Keras, Statistical Data Analysis)

PUBLICATIONS

J = Peer-reviewed journal article · C = Conference paper · B = Book chapter

Peer-Reviewed Journal Articles

[J1] Kriker, O., Bouchahda, N., **Mili, M.**, Sfar, R., Ben Abdallah, A., & Bedoui, M. H. (2026). Hybrid approach for automatic annotation and displacement estimation of cardiac fluid hemodynamics. *International Journal of Imaging Systems and Technology.*

[J2] Kechida, M., Slama, N., Kenani, M., **Mili, M.**, Salem, A., Krit, W., et al. (2026). Machine learning models in predictive factors for megaloblastic character of macrocytic anemia. *Leukemia Research Reports*, 100568.

[J3] Ben Khalifa, A., **Mili, M.**, Maatouk, M., Ben Abdallah, A., Abdellali, M., Gaied, S., et al. (2025). Deep transfer learning for classification of late gadolinium enhancement cardiac MRI images into myocardial infarction, myocarditis, and healthy classes. *Diagnostics*, 15(2), 207.

[J4] Salah, S., Jellad, A., **Mili, M.**, Chaabeni, A., Bedoui, M. H., & Ben Saad, H. (2025). Comparing efficacy and adherence of smartphone-guided exercises to conventional self-directed exercises for neck pain in office workers: A randomized controlled trial protocol. *PLoS One*, 20(9), e0329863.

[J5] Nouira, M., Yacoub, A., Abdallah, A. B., **Mili, M.**, Boudriga, H., Hmida, W., et al. (2025). L'IA peut-elle surseoir à la réalisation d'une scintigraphie osseuse dans le bilan initial d'un cancer de la prostate ? *Médecine Nucléaire*, 49(2), 128–129.

Conference Papers

[C1] Mili, M., Ben Abdeljelil, A., Ben Abdallah, A., & Bedoui, M. H. (2026). Weakly supervised ViT-conditioned text explanations in glioma histology. *IEEE International Joint Conference on Neural Networks (IJCNN 2026).*

[C2] Mili, M., Ben Abdallah, A., & Bedoui, M. H. (2026). NdLinear-attention based deep learning model for MGMT promoter methylation prediction in glioblastoma. *International Conference on Control, Decision and Information Technologies (CoDIT 2026).* Accepted.

[C3] Mili, M., Ben Abdeljelil, A., Manzanera, A., Ben Abdallah, A., Otero, J. J., & Bedoui, M. H. (2026). Quantitatively audited multi-view XAI for medical image classification: Application to MGMT promoter methylation. *International Conference on Agents and Artificial Intelligence (ICAART 2026).*

[C4] Messaoud, N. H., Zina, N., **Mili, M.**, Dhia, R. B., Abdallah, A. B., & Bedoui, M. H. (2025). Feature extraction through MS lesion and cerebral lobe segmentation for enhanced EDSS prediction. *International Symposium on Innovative Informatics of Biskra (IS-NIB 2025), IEEE.*

[C5] Mili, M., Abdallah, A. B., Kerkeni, A., Otero, J. J., & Bedoui, M. H. (2024). Approche ensembliste pour la prédiction du statut MGMT chez les patients atteints de GBM. *Applications Médicales de l'Informatique: Nouvelles Approches (AMINA 2024).*

[C6] Mili, M., Abdallah, A. B., Otero, J. J., Kerkeni, A., & Bedoui, M. H. (2023). Revolutionizing brain cancer diagnosis: Automated prediction of MGMT methylation status using histological images. *International Conference on Cyberworlds (CW 2023), IEEE*, 149–156.

[C7] Mili, M., Kerkeni, A., Abdallah, A. B., Otero, J. J., & Bedoui, M. H. (2022). Gradient boosting for predicting MGMT status in gliomas. *AMINA 2022.*

Book Chapters

[B1] Mili, M., Kerkeni, A., Ben Abdallah, A., & Bedoui, M. H. (2022). ICU mortality prediction using long short-term memory networks. *International Conference on Intelligent Data Engineering and Automated Learning (IDEAL 2022)*, 242–251. Cham: Springer International Publishing.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, R, SQL, XML/XSL, LaTeX, Ruby, MATLAB
- **Business Intelligence & Data Visualization: Microsoft Power BI**, Power Query, DAX, interactive dashboards, data modeling, data visualization

- **Deep Learning & Machine Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Theano, Caffe, XGBoost, LightGBM, Fastai, Microsoft Azure Machine Learning SDK
- **Explainable AI & Interpretability:** Weakly-supervised multiple-instance learning (MIL); Vision Transformer–based and attention-based architectures; quantitatively audited multi-view XAI frameworks; mechanistic interpretability methods for multimodal models
- **Medical Imaging & Biomedical Data:** Histopathology / whole-slide image (WSI) analysis; cardiac MRI classification; multimodal and multi-omic data fusion; electronic medical record (EMR) data processing
- **Databases:** MySQL, PostgreSQL, HBase, Elasticsearch, Oracle Database, MongoDB, Microsoft SQL Server
- **Web Development & Deployment:** HTML, CSS, JavaScript, Flask, Apache/Tomcat, Ngrok (lightweight model deployment for testing)
- **Languages:** Arabic (native), French (fluent), English (C1, TOEIC-certified 2019), Spanish (basic)

TEACHING & SUPERVISION

Contract Lecturer — ISIMM, Tunisia · 2024 – 2025

- Cloud & Virtualization (TP) — Software Engineering, 3rd-year License
- Mobile Development (TP) — Software Engineering, 3rd-year License

Contract Lecturer — Faculty of Sciences of Monastir (FSM) · 2023 – 2024

- SOA Architecture and Web Services (TP) — 3rd-year License
- Database Management Systems (TP) — 2nd-year License

Adjunct Lecturer — FSM · 2021 – 2023

- Data Science Introduction (Lecture + TD) — 1st- and 3rd-year License in Mathematics
- Statistical Programming with R (Lecture + TD) — 1st-year License in Mathematics

Student Project Supervision

- Design and Implementation of an Adaptable Diagnostic System Based on Clinical Data Analysis (Conception et réalisation d'un système de diagnostic adaptable basé sur l'analyse des données cliniques)
- Prediction of MGMT Gene Methylation Status from Histological Images Using Deep Learning (Prédiction du statut de méthylation du gène MGMT à partir d'images histologiques en utilisant l'apprentissage profond)
- Structuring and Analysis of Medical Data Using ML and DL Models : Application in Oncology (Structuration et analyse de données médicales par des modèles ML et DL : Application en oncologie)

Trainer, CEC-IA and Health Program — LTIM, Faculty of Medicine of Monastir · 2022, 2023

ACADEMIC SERVICE

- Session Chair, IEEE WCCI / IJCNN 2026 — Special Session “Trustworthy and Explainable Federated Learning: Towards Security and Privacy, Part II”
- Peer Reviewer — Computers in Biology and Medicine (IF 7.7, 2024); Journal of Machine Learning Research (2024); ICMLA (2021); AMINA (2022); International Conference on Electrical and Computer Engineering Researches (ICECER, 2026); Frontiers in Bioinformatics (2026)
- Organizing Committee — AMINA National Workshop (2022, 2026); 47th Congress of the Biomechanics Society (2022)
- Member, Tunisian Association for the Promotion of Applied Research (ATUPRA), since 2024

CONFERENCES & PROFESSIONAL DEVELOPMENT

Conference & Workshop Participation

- IEEE World Congress on Computational Intelligence / IJCNN 2026 — Maastricht, Netherlands (21–26 June 2026)
- IEEE International Conference on Cyberworlds (CW 2023) — Sousse, Tunisia (Oct 2022)
- International Conference on Intelligent Data Engineering and Automated Learning (IDEAL 2022)
- AMINA Workshop — Applications of Medical Informatics: New Approaches (2022, 2026) — Monastir, Tunisia
- 47th Congress of the Biomechanics Society (2022) — Monastir, Tunisia
- Digital Festival — Digital Technology and its Applications (2022) — Monastir, Tunisia
- STIC Schools: Artificial Intelligence and Medical Application / Medical Database Implementation (2019, 2021, 2023) — Monastir, Tunisia

Professional Development & Certifications

- Scientific Writing & Publishing: Wiley — “Step-by-Step Guide to Writing a Literature Review”; Wiley — “Scientific Writing Tips for Non-Native English Speakers”; LetPub — “Fighting Off Journal Rejections on the Path to Manuscript Acceptance” (2024); LetPub — “Building a Career Reputation via Peeref and Other Venues” (2023)
- Communication & Pedagogy: English Formation — Presentation (2023); Workshop — Web 2.0 Tools for Active Learning; 1st Pedagogical Congress; Article Writing and Bibliographic Management (2021)